

SYNOPSIS



On 16 July 2005 at 2220 (BST), in near darkness, two powerboats collided near Castle Point, St Mawes, resulting in the death of one of the helmsmen.

On the day of the accident, *Carrie Kate* was operated by a friend of the owner. This helmsman had held a 'quay-quay' licence for 2 years, issued by the Isles of Scilly Council, which entitled him to operate a vessel with fewer than 12 passengers between certain quays of the Isles of Scilly.

Carrie Kate was a 6.5m powerboat. She was propelled by a petrol inboard engine, via a stern-drive, enabling her to reach speeds of between 38-41 knots. There were no flares, lifejackets or VHF radio carried onboard, and her only navigation light was a red and green combined sidelight on her bow. The pole-mounted all-round white light, a non-permanent fitting which plugged into the starboard quarter, was broken and not onboard the vessel at the time of the accident.

The helmsman of *Kets* had been a fisherman for 11 years. Having grown up and fished out of St Mawes, he was very familiar with the local waters. He was one of four owners, who had bought *Kets* in April 2005.

Kets was a 4.55m Dell Quay dory, powered by an outboard engine which was controlled from a conning position on the starboard side. The vessel was capable of well over 20 knots. Again, no safety equipment was carried and the vessel had no navigation lights, as the owners did not intend to use *Kets* in the dark.

Carrie Kate visited several harbours in the Fal estuary on the day of the accident. The helmsman consumed alcohol throughout the day, and the St Mawes harbourmaster had noted the vessel due to the excessive wash she had created when departing the harbour earlier in the day. In the evening, *Carrie Kate* had again visited St Mawes, this time with four people onboard. At 2220 they departed St Mawes for the return trip to Falmouth. It is unclear whether the helmsman switched on his navigation light.

The helmsman of *Kets* had crossed to Falmouth from St Mawes, and from about 1800 onwards consumed alcohol. At 2215 the helmsman, his brother and a friend started their trip back to St Mawes. The sun had set at 2123, the weather was fine with little wind, and there was nearly a full moon.

At about 2220 those onboard *Kets* saw a bow wave and boat dead ahead. The helmsman turned *Kets* to port, but too late to avoid a collision with *Carrie Kate*. The helmsman of *Carrie Kate* had seen nothing but clear water ahead until *Carrie Kate* hit the starboard bow of *Kets*, riding over her and hitting the helmsman's console. The stern-drive of *Carrie Kate* was virtually sheared off as it hit the side of *Kets*. The helmsman of *Kets* was killed in the impact, his brother was flung into the water but fortunately remained conscious, and the other passenger remained in the boat but was knocked unconscious. Both vessels ended up stopped in the water.

Nobody was injured on board *Carrie Kate*, and they raised the alarm using a mobile phone. They were unable to assist the other vessel as they had no propulsion. The man in the water managed to swim to *Kets* and climb back aboard. The lifeboat arrived 10 minutes after the accident and the fatally injured helmsman was evacuated by helicopter shortly after.

Carrie Kate's helmsman was breathalysed after the accident and the alcohol in his body was found to be twice the legal limit permitted when driving a road vehicle. The helmsman of *Kets* was found to be over twice the legal limit also.

Analysis-

The tragic accident occurred because neither of the helmsmen saw the other vessel in enough time to take effective avoiding action. Contributing factors were:

- The speed of the vessels;
- The lack of navigation lights;
- The training and experience of the helmsmen; and
- That both helmsmen were navigating while under the influence of alcohol.

The COLREGS require that vessels proceed at a safe speed, taking due account of traffic density, visibility and the presence of background lights at night. They also require that appropriate navigation lights are exhibited between sunset and sunrise. This accident occurred an hour after sunset. It is thought highly unlikely that even the single navigation light on *Carrie Kate* was illuminated, thus both vessels were unlit. Although both helmsmen were described as experienced boatmen, their knowledge and application of the COLREGS was inadequate for operating high speed craft at night in a busy area of water.

In the UK, leisure craft users require no qualifications, although training is recommended. The RYA National Powerboat Scheme provides appropriate training at various levels where students learn the rudiments of the COLREGS and the benefits of carrying lifesaving equipment; both areas that were deficient in this case. How extensive this lack of essential knowledge is in the leisure industry, is difficult to quantify, as there is little statistical information collected on leisure accidents.

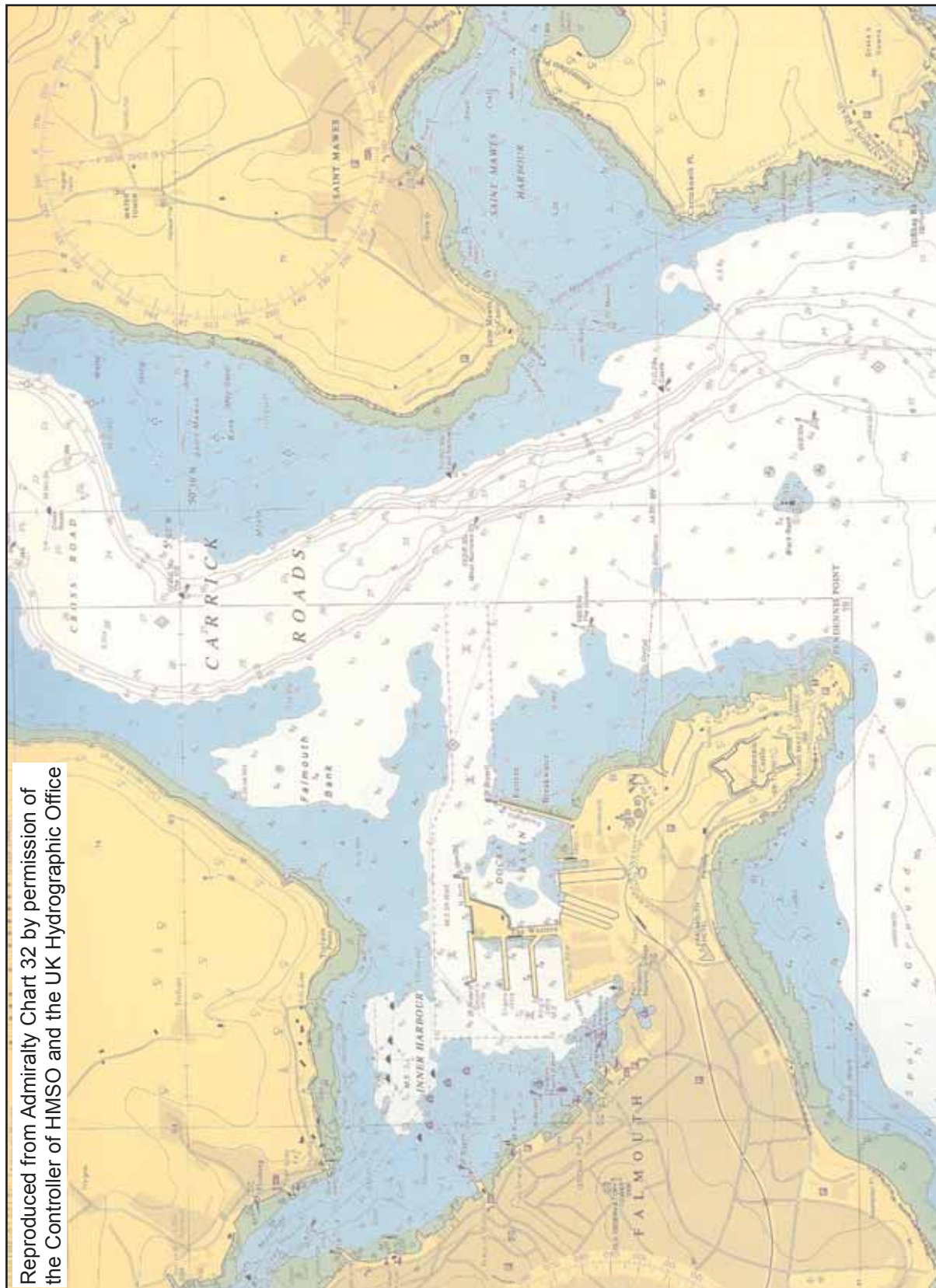
The key effects of alcohol consumption pertinent to this accident are: reduced peripheral vision, poor night vision, deterioration of judgment and slower reaction times. These factors are all key to navigating a high speed craft in near darkness. There are currently no national laws against recreational users navigating on water while under the influence of alcohol. Had there been such a law, both helmsmen might have thought twice before drinking and driving their water craft.

Local Fal estuary issues with respect to speed limits, unlit craft and bye-law approval were also raised during the investigation.

As a result of the recent MAIB investigation into the high speed grounding of the powerboat *Sea Snake*, which resulted in the loss of three lives, and this investigation, a recommendation has been made to the Department for Transport (DfT) concerning the

introduction of an alcohol limit for leisure users. The DfT is also recommended to improve the harbour bye-law approval process. The local harbour authorities have been recommended to review the standard of leisure boat activities in their area.

Chart 1



Reproduced from Admiralty Chart 32 by permission of the Controller of HMSO and the UK Hydrographic Office

Falmouth and St Mawes harbours

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *CARRIE KATE* AND ACCIDENT

Vessel details

Registered owner	:	Privately owned
Flag	:	UK
Type	:	Pleasure craft
Built	:	1998
Type	:	Sea Ray 215 Express Cruiser
Construction	:	GRP
Length overall	:	6.55m
Engine power and/or type	:	164kW (220HP), 5.0 litre Mercruiser
Maximum speed	:	38.3-41.7 knots (44-48 mph)
Other relevant info	:	Single stern drive

Accident details

Time and date	:	2225 16 July 2005
Location of incident	:	50° 09.2' N 005° 01.5'W
Persons on board	:	4
Damage	:	Stern drive disabled

1.2 PARTICULARS OF *KETS* AND ACCIDENT

Vessel details

Registered owner	:	Privately joint owned by 4 persons
Flag	:	UK
Type	:	Dell Quay Eurosport 15 Dory
Built	:	Pre 1970
Construction	:	GRP foam sandwich
Length overall	:	4.55m
Engine power and/or type	:	52kW (70hp) Yamaha outboard
Service speed	:	Estimate 20+ knots
Other relevant info	:	Console steering

Accident details

Time and date	:	As above
Location of incident	:	As above
Persons on board	:	3
Injuries/fatalities	:	1 fatality, helmsman - Ben Cochrane, 2 injured
Damage	:	Starboard side: significant impact damage above waterline

1.3 ENVIRONMENTAL CONDITIONS

All times are BST (UTC+1)

On Saturday 16 July 2005 the weather was fine and sunny. The visibility was good, there was only a gentle breeze, and it was ideal weather for boating. The evening was similarly good, with an almost full moon. Low water at Falmouth occurred at 1825, and high water was at 0108 the next day. It was neap tides. The sun set at 2123, and civil and nautical twilights occurred at 2205 and 2303 respectively.

1.4 NARRATIVE

1.4.1 Events leading to the collision

The helmsman of *Carrie Kate* on the day of the accident was an employee of the vessel's owner. The helmsman had been asked by his employer to house-sit while he was out of the country, and had been allowed the use of the employer's speedboat, *Carrie Kate* (**Figure 1**), which was moored at the bottom of his garden, on the Truro River.

Around midday on 16 July 2005, *Carrie Kate* left her mooring with her helmsman and two friends onboard. They proceeded south, and refuelled at Mylor before heading for St Mawes harbour (**Chart 1**). The harbourmaster met them as they tied up ready to visit a local public house and charged them the standard £2 mooring fee. Approximately an hour later, the helmsman and his friends left St Mawes harbour. The harbourmaster noticed *Carrie Kate* creating a large wash as she left, and made a mental note to speak to her helmsman when he saw him again. *Carrie Kate* made the short passage to Falmouth, and the helmsman's two friends were put ashore.

Not long after, the helmsman picked up another friend and his family in Falmouth, and suggested they go to St Mawes for lunch. *Carrie Kate*, with three adults and three children onboard, returned to St Mawes, arriving at 1410 and mooring against the harbour wall. As the helmsman tied up *Carrie Kate*, the harbourmaster approached to speak to him about the earlier wash incident. From the helmsman's demeanour, he assessed that he might be confrontational, and therefore handed him a St Mawes harbour guidelines pamphlet and suggested the helmsman read it over lunch. The harbourmaster did not charge the helmsman a second mooring fee.

Conscious of the tide and depth of water, *Carrie Kate's* helmsman left St Mawes harbour around 1630-1645. As she left, the harbourmaster again observed her generating excessive wash, which almost swamped a small dinghy heading out to the yacht moorings. The harbourmaster used his binoculars to note the name of the vessel, and considered calling ahead to see if someone could intercept her. However, he decided against this course of action as he did not know *Carrie Kate's* destination. The harbourmaster then closed the harbour

office and went home. *Carrie Kate* headed towards Mylor and followed a gig race back to Falmouth where the helmsman's friend and family left the vessel. The helmsman then secured *Carrie Kate* to the visitors' pontoon and went ashore.

Ben Cochrane was a part owner of *Kets* (**Figure 2**) and was her helmsman on the day of the accident. He had rowed out to the mooring in St Mawes harbour, and brought *Kets* back to shore at about 1630 ready to go to Falmouth for a working boat festival that was being held at a local public house. He and two friends made an uneventful crossing from St Mawes, and met Ben's brother in the public house at about 1730. A little later, *Carrie Kate*'s helmsman joined the four friends for no more than an hour, he being acquainted with them through local rugby and gig rowing.

At about 2115, the helmsman of *Carrie Kate* was in another public house to meet a male friend and two girls for a pre-arranged visit to St Mawes in *Carrie Kate*. He retrieved *Carrie Kate* from the visitors' pontoon and collected the three others at the Prince of Wales pier in Falmouth. The sun had just set, but it was still light and visibility was good. They arrived in St Mawes at about 2145 and went to a local hotel for another drink. Half an hour later, the helmsman was conscious that it was getting dark, and he was keen to return to the working boat festival in Falmouth.

The party departed St Mawes in *Carrie Kate* at about 2215. It is not known whether the navigation lights on *Carrie Kate* were switched on, but it is likely they were not, as the helmsman could not remember switching them on, and no witnesses could recall seeing navigation lights on *Carrie Kate* at the time of the accident. The helmsman headed south-west after clearing the quay and applied full throttle, as he had done all day. It is estimated that with four people onboard, this corresponded to a speed of 25-26 knots. He then turned the vessel in a gentle turn to starboard, and steered a straight course for the lights of Falmouth, clearing Castle Point to starboard. He was standing at the wheel to maximise his view ahead and, as it was a clear night with the moon out, the helmsman felt he had good visibility and believed he had clear water ahead on his course.

Ben Cochrane, his brother and two friends left the festivities at the first public house and had one drink at another local public house. Ben was conscious that it would be getting dark soon, so three of the group made their way back to *Kets*; the fourth person decided to stay in Falmouth. The group stopped briefly at a fast-food takeaway to allow Ben's brother to buy something to eat.

Some time after 2200, *Kets* left Falmouth with Ben at the wheel on the starboard side, his brother in the port side seat and their friend lying on his back in the bottom of the boat with his head towards the bow, this being the best position for a comfortable ride.

Within Falmouth harbour, the 8 knot speed limit was adhered to. Once clear of the limit, Ben increased to approximately $\frac{3}{4}$ throttle, as was his normal practice. This setting was sufficient to set the boat planing while delivering economical fuel consumption. The speed of *Kets* at this time was estimated to be 15-20 knots. No navigation lights were shown as *Kets* was not fitted with any. Castle Point and St Mawes Castle could be seen clearly, and Ben headed slightly to starboard of the point on a straight course with no craft visible ahead.

1.4.2 The collision

At about 2220, Ben and his brother saw the bow of a boat and the white of a bow wave dead ahead and very close. Ben turned *Kets* to port, but there was insufficient time to avoid a collision. Onboard *Carrie Kate*, the helmsman did not see *Kets*, although the passenger beside him saw a dark silhouette travelling from right to left briefly before the collision. There was a loud bang as *Carrie Kate*'s bow collided with the starboard side of *Kets*.

Those onboard *Carrie Kate* felt the vessel jump up as they hit *Kets* and passed over her, then came to a halt in the water on landing. The two people sitting on the aft seat were thrown into the air, but fortunately landed back on their seat. As *Carrie Kate* stopped, those onboard initially saw *Kets* outlined against St Mawes Castle to their left, apparently with no-one onboard, and one person in the water between them and *Kets* (**Chart 2**).

Onboard *Kets*, the force of the collision lifted Ben's brother out of his seat and threw him into the water. He was still conscious, and managed to swim to the water surface and tread water. The passenger lying down in the bottom of the boat was knocked unconscious. Tragically, Ben Cochrane was killed instantly in the impact.

1.4.3 Damage and post accident actions

Carrie Kate's stern drive demolished a large section of the starboard side of *Kets* (**Figures 3 & 4**), but the vessel remained afloat. The helmsman's console in *Kets* was knocked over the starboard side and was left hanging by the engine control cables. The engine had stopped as a result of the collision, but the kill-cord was still in place. *Carrie Kate*'s helmsman managed to start his vessel's engine, but he was unable to manoeuvre her to pick up the man in the water because the stern drive had been damaged in the impact and was hanging loosely from the transom (**Figure 5**).

There was a lot of shouting from those in the two boats, and also from guests at a wedding reception at St Mawes Castle who had seen the accident. One of the ladies on *Carrie Kate* raised the alarm for the emergency services, by calling the coastguard using her mobile telephone. A passing vessel offered some assistance, and provided a small set of flares to those onboard *Carrie Kate*. However, the battery of the only torch onboard died, preventing anyone from reading the instructions on the flares to make use of them.



Carrie Kate stern drive damage

The lifeboat crew were paged at 2226. The inshore lifeboat was launched at 2231 and arrived on scene by 2235, by which time the passing vessel had departed. The lifeboat put one crewman onboard *Kets* and then checked on *Carrie Kate*. On establishing that no one on *Carrie Kate* was injured, the lifeboat returned to *Kets* to administer first-aid. A pilot boat and the all-weather lifeboat arrived a short while later. Ben was transferred to the inshore lifeboat, and from there was evacuated by helicopter. The other two injured crew from *Kets* were taken into St Mawes to a waiting ambulance and were taken to hospital. Ben's brother required an operation on a deep laceration on his right arm, and their friend suffered significant bruising and some internal injuries.

A search was briefly conducted for a fourth person, believed to be from *Kets*, until it was confirmed that only three people had been onboard when she left Falmouth.

Carrie Kate's passengers were transferred to the lifeboat and the vessel was towed into St Mawes harbour.

The police briefly interviewed those onboard and the helmsman was breathalysed. The alcohol in his body was found to be twice the legal limit permitted when driving a road vehicle. The postmortem for Ben Cochrane also revealed he was over twice the legal limit for a road vehicle.

The helmsman's console, with its small windshield, was positioned on the starboard side of the boat, the throttle being mounted just inside the starboard grab-rail. A kill-cord was fitted, but the helmsman was not wearing it at the time of the accident. *Kets* had no navigation lights as the owners had not intended that she would be used in the dark. There were no other instruments or safety equipment onboard.

Kets was bought early in 2005 by four men in equal partnership. Ben Cochrane and his father were two of the owners. The previous owner was a friend of the Cochrane family and, as the vessel had been based in St Mawes for many years, she was well known to her new owners who kept the boat on a mooring on the Percuil River, not far from St Mawes harbour. *Kets* was described as being tidy and in good condition, and was intended purely for recreational activities in the local waters and not for use beyond St Anthony's lighthouse.

1.7 LOCAL INFORMATION

1.7.1 General

The Fal estuary (**Chart 3**) is a large natural harbour, and is a busy port as well as a popular recreational area. As a result of its diverse marine wildlife, the area has also been designated a special area of conservation (SAC). There are believed to be 6,500 boat moorings in the area, including those on the Helford River, and this figure does not include boats dry stored or launched from slipways. During the summer months in particular, the number of craft on the water is considerable. The area is well served by public houses and restaurants adjacent or near to the water.

The 2.3 nm route between Falmouth and St Mawes is relatively sheltered and is especially busy during the summer season. At the entrance to St Mawes harbour area, an unlit cardinal buoy indicates safe passage to the south of Lugo rock. However, many small craft pass north of this mark as it is a shorter route and has adequate water at most states of the tide. At night, there is a reasonable amount of background light from Falmouth, but not from St Mawes. During the investigation it was reported on several occasions to MAIB inspectors that significant numbers of unlit craft were on the water at night during the summer, some of which were believed to be angling or vessels engaged in illegal fishing.

The control of the waters in this region, for mostly historical reasons, is complex. **Chart 3** shows the areas of responsibility of Falmouth Harbour Commissioners, Falmouth Docks, Carrick District Council (including Truro Harbourmaster) and the St Mawes Pier and Harbour Company.

1.8 THE INTERNATIONAL REGULATIONS FOR PREVENTING COLLISIONS AT SEA (COLREGS)

The COLREGS apply to all vessels on the high seas, and in all waters connected to the high seas navigable by sea-going vessels. As such, a basic understanding of the COLREGS is essential before venturing on to the water.

Annex D contains extracts from the COLREGS which are pertinent to this accident.

1.9 SECOND COLLISION NEAR CASTLE POINT

A second collision near Castle Point occurred on 12 August 2005 between a water taxi and an angling vessel. The angling vessel's bow was severed in the collision, and the boat sank. The two occupants were rescued by the water taxi, which suffered some minor damage. Fortunately there were no injuries.

Accident investigations were carried out by the St Mawes Harbourmaster and Truro Harbourmaster. The former, because the accident was in St Mawes harbour waters, and the latter because Truro Harbourmaster had licensed the helmsman of the water taxi.

The accident occurred at night, very close to the location of the *Carrie Kate* and *Kets* collision. The water taxi had left St Mawes with just the helmsman onboard, and was headed towards Mylor. As he rounded Castle Point, the helmsman saw an oyster punt immediately ahead. He turned to starboard, but the water taxi still caught the bow of the punt. The helmsman estimated he was doing 20-24 knots at the time of the collision. The men in the punt had been fishing, and were not showing any fixed navigation lights.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 CAUSE OF ACCIDENT

The cause of this tragic accident was that neither of the vessels involved saw the other in enough time to take effective avoiding action. Contributing to this was:

- The speed of the vessels;
- The lack of navigation lights;
- The training and experience of the helmsmen; and
- That both helmsmen were navigating while under the influence of alcohol.

2.3 FATIGUE

Fatigue is not considered to be a contributory factor in this accident.

2.4 COLREGS FACTORS

2.4.1 Speed

High closing speed was a significant factor in this accident. A slower closing speed would have increased the time available to sight the other vessel and react. It would also have lessened the impact, and therefore the effect of the accident. The resulting consequences of the other contributory factors would therefore have been diminished.

Safe speed, as defined in Rule 6 of the COLREGS (**Annex D**), requires that *'Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions'*. Rule 6 discusses the factors to be taken into account, specifically: visibility, traffic density and the presence of background lights. The helmsmen should have considered the following:

- As local water users, they might well have been aware that unlit boats were a regular hazard.
- Although it was not quite fully dark, visibility of unlit objects was significantly reduced when compared to daylight visibility. The background light from the shore would have hampered further the visibility of unlit objects, and the area was known to be busy, particularly in the summer.

- Although under normal conditions both helmsmen could reasonably have expected other vessels operating at night to be showing identifying navigation lights, their own vessels were almost certainly unlit.

Neither helmsman believed they were travelling too fast for the conditions, and both thought they had clear water ahead of them. However, their vision was not sufficiently good so as to prevent them from colliding at high speed with an unlit object, in this case another boat. Their speed was therefore too fast for the prevailing conditions.

Excessive speed can also affect the safety of other water users. Operators of powerboats can forget the hazard their wash creates, particularly to smaller craft. Often, failure to realise there is a problem is due to the effect of the wash being felt some distance from the vessel creating it. Waves created by wash can travel long distances, and the waves will increase in size if they pass from deep water into shallower areas such as beaches, anchorages and harbours. Inexperienced boat users may be unaware of the hazard created by excessive wash if they have not undergone any training before taking to the water. However, experienced boat operators will be aware of the dangers of excessive wash and most, in consideration of other water users, will moderate their speed accordingly.

2.4.2 Rights of way

The COLREGS (**Annex D**) also define, for different situations, which vessel must give way and what avoiding action they must take to prevent a collision. When two vessels are approaching each other head on or near head on Rule 14 applies, and both vessels shall alter course to starboard. If two vessels are in a crossing situation then Rule 15 applies, and the vessel with another on her starboard side shall keep clear.

A number of factors were considered relevant to this accident:

- a) Given that *Carrie Kate* and *Kets* were on reciprocal passages between St Mawes and Falmouth harbours it is highly likely they were in a head on, or near head on situation. Instinctively, in this situation, both helmsmen should turn to starboard and not to port, which is what *Kets* did on this occasion, after seeing a bow wave broadly ahead. This may suggest the vessels were not head on, or that the rules in the COLREGS were not appreciated.
- b) A passenger in *Carrie Kate* believes they saw a dark silhouette moving from right to left immediately before the collision. This, coupled with the fact that *Kets* turned to port may suggest there was some crossing component to the two vessels' courses. If the dark silhouette moving from right to left was *Kets*, then *Carrie Kate* was the give way vessel and should have taken avoiding action.

- c) Evidence from the impact damage shows that *Carrie Kate* struck *Kets* at an acute angle on the starboard side, and crossed *Kets* at an angle of 20°-30°. This may suggest that *Kets* had *Carrie Kate* on her starboard side and that *Kets* was the give way vessel.

MAIB was unable to be certain of the exact approach angles, and whether either vessel bore more burden than the other for altering course. It was also impossible to determine whether, if *Kets* had altered to starboard instead of port, she would have avoided the collision. However, the argument of which vessel should or should not have given way is academic, as neither vessel's helmsman saw the other, in enough time to prevent a collision.

2.4.3 Navigation lights

Part C of the COLREGS (**Annex D**) requires the appropriate navigation lights to be exhibited from sunset to sunrise [Rule 20], and vessels of under 12m to exhibit an all-round white light and sidelights [Rule 23]. Navigation lights achieve three main purposes: they make a vessel visible at night; they help the observer determine the aspect of the vessel; and they provide vital information on the type of vessel and dangers it presents. In this accident, the conditions were described as being not really dark, due to the moonlight. However, the COLREGS leave no room for ambiguity – sunset had occurred an hour before the accident, and both vessels should have been showing the appropriate navigation lights.

Kets was not equipped with any navigation lights as her owners did not intend to use her at night. It is assessed, therefore, that in choosing to navigate after sunset, without navigation lights, the helmsman was either unaware of the requirements in the COLREGS, or chose to ignore them.

Carrie Kate could only have exhibited her combined red and green sidelights, the all-round white light pole fixture being absent from the vessel at the time of the collision. Witnesses ashore and in *Kets* could not remember seeing any navigation lights on *Carrie Kate*. Also, given the helmsman's lack of experience of navigating at night, and other evidence collected, it is concluded that the sidelights on *Carrie Kate* were probably not illuminated. Like the helmsman of *Kets*, the helmsman was either unaware of the requirements of the COLREGS, or chose to ignore them.

2.4.4 COLREGS awareness

There is no requirement for leisure craft users, or fishermen operating fishing vessels under 16.5m in length, to undertake any training in, or demonstrate any knowledge of the COLREGS, though they are bound by the regulations themselves to apply them when navigating on the high seas or waters connected to the high seas.

2.8.4 Culture

The sheltered and picturesque Fal estuary, with its many creeks, is an ideal area for recreational craft. It is justifiably popular with the leisure boating community. There is also a significant number of restaurants and public houses in close proximity to the water. Three separate parties in this investigation apparently saw nothing wrong in consuming alcohol and navigating on the water. This might be indicative of a wider drinking and boating culture in this area. The local bye-laws currently do little to discourage this activity as, in their current form, they are unenforceable. It would be appropriate for the issue to be the subject of a local education campaign, which should be aimed at persuading the leisure boating community that drinking and boating is dangerous, and can kill.

2.9 SIMILAR ACCIDENT

A month after the fatal accident, a second collision occurred in the same area off Castle Point between a high speed water taxi and an unlit fishing dory. The dory was cut in two by the water taxi and sank, but fortunately both the occupants of the fishing vessel were uninjured. In this accident, alcohol was not involved, but in other respects there were many striking similarities between the two accidents. Speed, and the absence of any navigation lights on the angling vessel, were again significant issues, preventing the helmsman of the water taxi from taking evasive action early enough to avoid the collision. This second accident serves to reinforce many of the safety issues identified in this investigation, and the need for local action.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been highlighted by the investigation. They are not listed in any order of priority.

COLREGS factors

1. The closing speed of the vessels was a key factor in the collision. [2.4.1]
2. Neither vessel was using navigational lights as required by the collision regulations. [2.4.3]
3. Both skippers were described as experienced boatmen yet, in this accident, their application of the COLREGS was inadequate. [2.4.4]

Training

4. Neither craft involved in the accident carried any useful safety equipment. [2.5]
5. The engine's kill-cord was not connected to the helmsman in *Kets*. [2.5]
6. Leisure craft users are not required to hold any qualification in vessel operation before taking their boats to sea. [2.6]
7. Neither the mandatory fishing safety courses, nor the Isles of Scilly boatman's qualifications provided the helmsmen of the two vessels with the level of training that was sufficient for them to operate high speed craft at night. [2.6]
8. The extent to which leisure boat users operate their craft without appropriate levels of safety equipment, or with insufficient knowledge of the principles of navigation and the COLREGS, is difficult to quantify because there is little reliable statistical data collected on leisure accidents. [2.6]

Influence of alcohol

9. Both helmsmen were operating their vessels while under the influence of alcohol, and were over twice the drink driving limit for road vehicles. [2.7.2]
10. There is currently no national law against navigating a privately owned leisure vessel while under the influence of alcohol. [2.7.3]

Local issues

11. There was no speed limit within the waters of St Mawes Harbour. [2.8.1]
12. Unlit craft frequently operate at night in areas of the Fal estuary. [2.8.2]
13. The effectiveness of some local harbour bye-laws is questionable. [2.8.3]
14. Evidence collected in this investigation might be indicative of a wider drinking and boating culture in the Fal estuary. [2.8.4]